

CYBERSECURITY-LABS

INFORMATION SECURITY

LAB REPORT



Práctica: Encryption Ubuntu Server

Informe de práctica

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Information Security

Cybersecurity / Cloud / Networks / GRC



Paso 1: Para empezar, utilizo el comando echo para crear un archivo llamado documento.txt con el texto "esto es un archivo de prueba SO3 V.9". Luego, procedo a cifrar este archivo de forma simétrica utilizando el algoritmo AES256 con la herramienta gpg, generando el archivo cifrado documento.txt.

```

Memory usage: 194      IPv4 address for enp3s1: 192.168.66.102
Swap usage:  0A

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/can or run: sudo apt status

*** System restart required ***
last login: Wed Oct 29 02:14:45 2025 from 192.168.66.102
victor_2025066@victor-server:~$ echo "Esto es un archivo de prueba SOY V.G" > documento.txt
victor_2025066@victor-server:~$ gpg --symmetric --cipher-algo AES256 documento.txt

I

Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/
Processing triggers for ufw (0.36-2-6) ...
Processing triggers for man-db (2.12.0-4b14d2) ...
Processing triggers for libc-bin (2.39-0ubuntu6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Scanning kernel version:
6.8.0-85-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-85-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should do
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/bus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service
No containers need to be restarted.

User sessions running outdated binaries:
victor_2025066@0 session 11: login(943)
victor_2025066@0 user-manager.service: systemd[1160]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_2025066@victor-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link-localhex 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host
        valid_lft forever preferred_lft forever
    inet6 ::1::1/128 scope host no_mcast
        valid_lft forever preferred_lft forever
2: enp3s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link-ether 08:00:27:78:94:61 brd ff:ff:ff:ff:ff:ff
    inet 192.168.66.129/24 metric 100 brd 192.168.66.255 scope global dynamic enp3s1
        valid_lft 6665sec preferred_lft 6665sec
    inet6 fd01:a00:27ff:fc79::61:64 scope link
        valid_lft forever preferred_lft forever
victor_2025066@victor-server:~$ [ 1644.576793] c1000 0000:00:03:0 enp3s1: Reset adapter

```

Paso 2: El sistema me solicita que ingrese y reconfirme una frase de contraseña (Passphrase) que se utilizará como clave para el cifrado AES256. Introduzco una clave segura en ambos campos, asegurando que coincidan, para que la herramienta gpg pueda completar el proceso y proteger la información del archivo.

```

Created symlink /etc/systemd/system/sshd.service.requires/socket + /usr/lib/systemd/system/socket.
Processing triggers for systemd (0-96.38-4) ...
Processing triggers for man-db (2.12.0-3ubuntu1) ...
Processing triggers for libnss-bin (2.39-0ubuntu1.6) ...
Scanning processes...
Scanning candidate...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.8.0-35-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-36-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
systemd-restart-systemd.service
systemctl restart systemd-logind.service
systemctl restart snmited-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
victor_20250604 $ session id: login[93]
victor_20250604 $ user manager service: systemd[160]

No VM guests are running outdated hypervisor (qemu) binaries on this host.

victor_20250604#victor-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qlen 1000 state UNKNOWN group default qlen 1000
    link-loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1::1 scope host noprefixroute
            valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qlen: fq_codel state UP group default qlen 1000
    link-ether 00:00:27:79:04:01 brd ff:ff:ff:ff:ff:ff
    inet 132.160.40.129/24 metric 100 brd 132.160.40.255 scope global dynamic enp0s3
        valid_lft 6055sec preferred_lft 6055sec
    inet6 fe80::a80:27ff:fe79:a104 scope link
        valid_lft forever preferred_lft forever
victor_20250604#victor-server:~$ [ 1644.576791] c1000 0000:00:03:0 enp0s3: Reset adapter

```

Paso 3: Una vez que he ingresado la contraseña, la herramienta gpg me confirma que ha creado el archivo cifrado y me muestra su ubicación en la misma carpeta (documento.txt.enc). También me notifica que ha creado automáticamente el directorio .gnupg en mi directorio personal, el cual utilizará para almacenar las claves y configuraciones.

```
Victor_2025064@victor-server:~$ echo "Esto es un archivo de prueba S03 V.3" > documento.txt
Victor_2025064@victor-server:~$ gpg --symmetric --cipher-algo AES256 documento.txt
gpg: directory '/home/victor_2025064/.gnupg' created
gpg: keybox '/home/victor_2025064/.gnupg/pubring.kbx' created
Victor_2025064@victor-server:~$

Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket + /usr/lib/systemd/system/ssh.socket.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4b142) ...
Processing triggers for libc-bin (2.39-0ubuntu6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade?
Running kernel version:
6.8.0-85-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
systemctl restart /dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
victor_2025064 @ session #1: login(943)
victor_2025064 @ user manager service: systemd(1160)

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_2025064@victor-server:~$ ip a
1: lo: C00P0AC0:UP:LOWER_UP:mtu 65536 qlen 1000 scope state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
```

Paso 4: Ahora, ejecuto el comando gpg --gen-key para crear mi par de claves (pública y privada). Primero, selecciono el tipo de clave que deseo (opción 1, RSA y RSA), luego defino el tamaño de la clave en 4096 bits y establezco que la clave no debe expirar, eligiendo '0' para que sea permanente.

```
Deep sleep: OK

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run sudo apt update

*** System restart required ***
Last login: Wed Oct 29 02:14:45 2025 from 192.168.48.102
victor_2025064@victor-server:~$ echo "Esto es un archivo de prueba S03 V.3" > documento.txt
victor_2025064@victor-server:~$ gpg --symmetric --cipher-algo AES256 documento.txt
gpg: directory '/home/victor_2025064/.gnupg' created
gpg: keybox '/home/victor_2025064/.gnupg/pubring.kbx' created
victor_2025064@victor-server:~$ gpg --full-generate-key
gpg (GnuPG) 2.4.4: Copyright (C) 2024 g10, code 0x00
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
(1) RSA and RSA
(2) RSA and ElGamal
(3) RSA (sign only)
(4) RSA (sign only)
(5) ECC (sign and encrypt) "default"
(10) ECC (sign only)
(14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (2048) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
0 = key does not expire
<nn> = key expires in n days
<nnm> = key expires in n months
<nnY> = key expires in n years
key = key expires at time
key does not expire at all
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: victor
Email address: victor@esample.com
Comment: lol
You selected this USER ID:
"victor (lol) <victor@esample.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? Q
```

Paso 5: Después de configurar los parámetros de mi clave GPG, el sistema me pide que ingrese una frase de contraseña (Passphrase) para proteger mi nueva clave privada. Ingreso una clave fuerte y segura, ya que esta frase es fundamental para cifrar y descifrar archivos, garantizando que solo yo pueda usar la clave.

```
logn...
x Please enter the passphrase to protect your new key
x 1
x Passphrase: *****
x
x <OK> <Cancel>
x

Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket + /usr/lib/systemd/system/ssh.socket.
Processing triggers for nfs (4.3.0-2.6) ...
Processing triggers for nfs-kernel-server (2.3.9-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.8.0-55-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-56-generic.
Restarting the system to load the new kernel will not be handled automatically, so you should consider restarting the system manually.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/usb.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

Other sessions running outdated binaries:
victor_20250664 # session #1: login(943)
victor_20250664 # user manager service: systemd(1160)

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@victor-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc mq master state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0c3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:79:04:c1 brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.129/24 metric 100 brd 192.168.68.255 scope global dynamic enp0c3
        valid_lft 6665sec preferred_lft 6665sec
    inet6 fe80::a00:27ff:fe79:40c1/64 scope link
        valid_lft forever preferred_lft forever
victor_20250664@victor-server:~$ ifconfig
victor_20250664@victor-server:~$
```

Paso 6: Después de introducir mis datos (nombre, email y comentario) para el ID de mi clave, el sistema me solicita generar entropía moviendo el ratón y presionando teclas. Finalmente, confirma que mi clave pública y privada han sido generadas y almacenadas con éxito, mostrando el ID único de la nueva clave GPG creada.

```
(Please select what kind of key you want:
(1) RSA and RSA
(2) RSA and ElGamal
(3) RSA (sign only)
(4) RSA (sign only)
(5) ECC (sign and encrypt) "default"
(10) ECC (sign only)
(16) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
0 = key does not expire
<ok> = key expires in n days
<ok> = key expires in n weeks
<ok> = key expires in n months
<ok> = key expires in n years
Key is valid for? (0) 0
Key does not expire at all
Is this correct? (y/n) y

GnuPG needs to construct a user ID to identify your key.

Real name: victor
Email address: victor@example.com
Comment: lol
You selected this USER ID:
"victor (lol) <victor@example.com>"

Change (N)ame, (C)omment, (E)mail or (O)key/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disk) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disk) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.

gpg: /home/victor_20250664/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/victor_20250664/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/victor_20250664/.gnupg/openpgp-revocs.d/F3F75D8E848992A203C462F8487013
60D58C48.rev'
Public and secret key created and signed.

pub rsa4096 2025-10-29 [SC]
    F3F75D8E848992A203C462F848701360D58C48
uid victor (lol) <victor@example.com>
exp rsa4096 2025-10-29 [E]

victor_20250664@victor-server:~$
```

Paso 7: Utilizo el comando `gpg --encrypt --recipient "victor (lol) <victor@example.com>" documento.txt` utilizando el ID de mi propia clave GPG recién creada para cifrar el archivo `documento.txt`. El sistema me avisa que el archivo de salida con extensión `.gpg` ya existe (del cifrado simétrico anterior) y confirmo que deseo sobrescribirlo con el nuevo cifrado asimétrico.

```
victor_20250664@victor-server:~$ gpg --encrypt --recipient "victor (lol) <victor@example.com>" documento.txt
File 'documento.txt.gpg' exists. Overwrite? (y/N) y
victor_20250664@victor-server:~$
```

Paso 8: Primero, revisé el estado del cortafuegos UFW y me aseguré de que estuviera activo (active) si estaba apagado. Luego, procedí a activar las reglas para los servicios esenciales requeridos en la práctica: permití el tráfico entrante para SSH (22), Web (80/443) y FTP (21), asegurando la persistencia de estas reglas tras un reinicio.

```
Victor_0225066@Victor-server:~$ sudo ufw status verbose
UFW status:
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
21/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
21/tcp (v6) ALLOW IN Anywhere (v6)
```

```
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket + /usr/lib/systemd/system/ssh.socket.
Processing triggers for ufw (0.36-2.4) ...
Processing triggers for man-db (2.12.0-4b1b162) ...
Processing triggers for libc-bin (2.39-0ubuntu6.6) ...
Scanning processes...
Scanning candidates...
Scanning Linux images...

Sending kernel upgrade!
Running kernel version:
6.8.0-85-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.
Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.
Restarting services...
```